Comprehensive Road Scene Understanding for Autonomous Driving

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Abstract

Road scene understanding is a fundamental challenge in computer vision, requiring models to segment individual objects, recognize background regions, and detect out-of-distribution (OOD) anomalies in safety-critical environments. This work evaluates segmentation and anomaly detection on urban driving scenes, comparing a per-pixel baseline (ERFNet [14]) with EoMT [9], a mask classification model built on a DINOv2 pre-trained backbone. Starting from a COCO pre-trained checkpoint, we fine-tune EoMT on Cityscapes under three strategies: head-only training, partial backbone unfreezing (blocks 8–11), and parameter-efficient fine-tuning via LoRA [8]. LoRA achieves the best mIoU of 75.54% with only 7M trainable parameters, outperforming partial unfreezing (74.31%, 36M parameters). Finally, we evaluate anomaly segmentation performance across five benchmarks using post-hoc scoring methods (MSP, MaxLogit, MaxEntropy) for both architectures and RbA [12] exclusively to EoMT, with mask-based models reaching up to 94.05 AuPRC on SMIYC RO-21 versus 4.63 for ERFNet. Link to the project's GitHub repository

1. Introduction

Safe autonomous driving depends on the ability to accurately parse road scenes, not only recognizing known objects such as vehicles and pedestrians, but also detecting unfamiliar obstacles that fall outside any predefined category. Achieving high-confidence object segmentation and anomaly detection is inherently complex due to the extreme variability of driving environments, ambiguous object boundaries, and frequent occlusions. Optimal scene comprehension requires distinguishing between continuous background elements ("stuff") and discrete, countable objects ("things"). While semantic segmentation classifies every pixel, effectively merging objects of the same class, instance segmentation isolates individual objects while discarding the background. Panoptic segmentation [10] resolves this dichotomy. By unifying both approaches, it assigns a class label to every pixel and a unique instance identifier to every discrete object, yielding a complete and unified representation of the scene.

1.1. The Shift Toward Anomaly Segmentation

Despite the significant leap forward provided by panoptic segmentation, its reliance on a closed-set assumption remains a critical limitation. Closed-set models assume the operational world is strictly defined by the training data. Because softmax normalizes confidence over a fixed set of classes, and sigmoid heads can only activate for known categories, models become prone to overconfident misclassifications when encountering novel objects in dynamic driving environments. This limitation necessitated a shift in objectives: rather than exclusively asking what every known object in an image is (panoptic segmentation), models must also determine if there are any elements present that do not belong to the established taxonomy (anomaly or OOD segmentation). Early per-pixel baselines like ERFNet [14] were structurally limited in detecting anomalies. The mask-classification paradigm (MaskFormer [3], Mask2Former [4]) evolved into EoMT [9], which further simplifies this via an encoder-only architecture injecting queries directly into a DINOv2 [13] Vision Transformer backbone, while RbA [12] introduced region-level strategies for isolating unknown objects. This work evaluates EoMT in a zero-shot setting on Cityscapes via a structured COCO-to-Cityscapes class mapping [5, 11], fine-tunes it under three strategies (head-only, partial backbone unfreezing, and LoRA [8]), and benchmarks anomaly segmentation performance of both EoMT and ERFNet using post-hoc scoring methods (MSP, MaxLogit, MaxEntropy, RbA) across five datasets.

2. Methodology

2.1. Architectural Overview and Preliminary Setup

ERFNet [14] serves as our per-pixel segmentation baseline, producing dense predictions without any explicit notion of object-level structure, a limitation that leads to false positives at semantic boundaries and noisy anomaly maps (Section 2.4). EoMT [9] represents the evolution of the mask classification paradigm introduced by MaskFormer and Mask2Former. Cityscapes [5] (2,975 training / 500 validation images, 19 classes) serves as the target domain; COCO [11] (133 panoptic categories) serves as the source domain for the upstream checkpoint. We evaluate two official EoMT checkpoints: EoMT-COCO, whose zero-shot performance on Cityscapes defines the lower bound, and EoMT-Cityscapes, which represents the upper bound targeted by fine-tuning. Anomaly segmentation is evaluated on five benchmarks [1, 2]: SMIYC RA-21 (large objects), SMIYC RO-21 and FS Lost&Found (small, distant obstacles), FS Static (OOD objects superimposed on Cityscapes scenes), and Road Anomaly (high domain shift from Cityscapes).

2.2. Phase 1: Zero-Shot Evaluation and Cross-Dataset Mapping

To establish baseline performance, we evaluate both EoMT-COCO and EoMT-Cityscapes on the Cityscapes validation set without any fine-tuning. We define a structured COCO $\rightarrow$ Cityscapes mapping that handles three cases:

(i) **many-to-one**, where multiple COCO categories merge into a single Cityscapes class (e.g., potted-plant, tree-merged, flower $\rightarrow$ vegetation);

(ii) **one-to-one**, for classes with direct equivalence (e.g., car $\rightarrow$ car, person $\rightarrow$ person);

(iii) **no mapping**, where COCO categories with no Cityscapes equivalent (e.g., laptop, pizza) are assigned the ignore index (255) and excluded from evaluation.

The rider class, absent from COCO, is treated as unmapped.

2.3. Phase 2: Downstream Fine-Tuning Strategies

Starting from EoMT-COCO, we explore three fine-tuning strategies of increasing adaptation scope to leverage its rich visual understanding on the domain-specific Cityscapes dataset.

2.3.1. Strategy A: Head-Only Fine-Tuning

The ViT-B backbone is kept fully frozen and only the prediction heads are updated: the classification head (`class_head`), the mask prediction MLP (`mask_head`), the upsampling module (`upscale`), and the learnable query token embeddings (`network.q`), amounting to $\sim$6.7M trainable parameters out of 84M total (8.0%).

2.3.2. Strategy B: Partial Backbone Unfreezing (Blocks 8–11)

Building on Strategy A, we additionally unfreeze the four deepest transformer blocks (blocks 8–11), with blocks 0–7 remaining frozen. This increases trainable parameters to $\sim$36M (43%), introducing a non-negligible risk of overfitting on the Cityscapes training images, which motivates Strategy C.

2.3.3. Strategy C: Parameter-Efficient Fine-Tuning via LoRA

We apply Low-Rank Adaptation (LoRA) [8] to combined QKV projection of each attention layer. For each target linear layer W_{orig}, LoRA introduces $A \in \mathbb{R}^{r\times d}$ and $B \in \mathbb{R}^{d\times r}$ (r=8, α=16), adding $\Delta W = B \cdot A$ scaled by α/r, while keeping W_{orig} frozen. Head components are unfrozen identically to Strategies A and B, yielding $\sim$7M trainable parameters ($\sim$294K LoRA matrices) out of 84M total (8.3%).

Training Configuration

All strategies use AdamW (lr $= 10^{-4}$, LLRD $= 0.8$, weight decay $= 0.05$), polynomial LR scheduling, and train for 8 epochs on a single NVIDIA T4 with batch size 1 and fp16 mixed precision.

2.4. Phase 3: Anomaly Segmentation

This phase addresses the detection of OOD pixels that do not belong to any predefined road-scene category. Anomaly detection is formulated as a per-pixel scoring task, where each pixel receives a continuous score proportional to its likelihood of being an anomaly.

2.4.1. Evaluation Metrics

Performance is measured by two threshold-free metrics: AuPRC, which summarizes anomaly score ranking quality and is robust to the severe class imbalance between anomalous and in-distribution pixels; and FPR$_{95}$, the fraction of in-distribution pixels misclassified when 95% of anomalous pixels are correctly identified. Images with no anomalous pixels are excluded from evaluation.

2.4.2. Pixel-Based Baselines on ERFNet

A pre-trained frozen ERFNet baseline with C=20 output channels executes a single forward pass per image. From the resulting logits tensor $[C, H, W]$, three post-hoc scores are computed simultaneously: MaxLogit [7], MSP and MaxEntropy.

2.4.3. Mask-Based Baselines on EoMT and the RbA Method

To evaluate a mask-classification approach, we adapt the EoMT architecture using a pre-trained DINOv2 encoder backbone. Instead of outputting standard per-pixel logits, EoMT uses a set of learned object queries to predict binary masks and corresponding class distributions. Following [12], a composite pixel-level map $L(c)$ is reconstructed by aggregating these queries:

$$L(c) = \sum_i p_i(c) \cdot \sigma(m_i), \tag{1}$$

where $p_i(c)$ denotes the post-softmax class probability for query i, and $\sigma(m_i)$ represents its post-sigmoid binary mask probability. $L(c)$ is thus a weighted sum aggregating softmax class probabilities scaled by sigmoid mask activations across all queries. Although bounded by the total number of queries, $L(c)$ serves as a proxy logit tensor, allowing the direct computation of comparable MSP, MaxLogit, and MaxEntropy scores. Furthermore, this mask-based formulation enables the utilization of the Rejected by All (RbA) method, defined as:

$$\text{RbA} = -\sum_c \sigma(L(c)), \tag{2}$$

where each class channel acts as a one-vs-all classifier, flagging a pixel as anomalous when $\sigma(L(c))$ remains low across all known categories c. Prior to scoring, EoMT-COCO logits are remapped to the 19 Cityscapes classes via the fixed mapping defined in Section 2.2, discarding COCO-only categories and aligning the inference space across all models.

2.4.4. Temperature Scaling

To evaluate the impact of logit calibration, temperature scaling [6] is applied to the MSP scores of the EoMT checkpoints. The proxy logits are scaled by a positive scalar T before the softmax operation:

$$\text{MSP}(c, T) = 1 - \max_{c} \text{softmax}\left(\frac{L(c)}{T}\right)_c, \tag{3}$$

adjusting the entropy of the probability distribution without altering the predicted winning class. A grid search is performed over $T \in \{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$, reporting both the uncalibrated baseline ($T = 1.0$) and the optimal configuration.

3. Experimental results

3.1. Zero-Shot Evaluation

Table 2 reports the mIoU of EoMT-COCO and EoMT-Cityscapes on the Cityscapes validation set. EoMT-Cityscapes achieves 81.68% mIoU, while EoMT-COCO reaches 52.31% after class mapping. Figure 1 provides a qualitative comparison: EoMT-Cityscapes produces segmentations closely aligned with the ground truth, while EoMT-COCO shows significant confusion in domain-specific regions. Notable per-class results for EoMT-COCO include high IoU for road (0.94), sky (0.90), and car (0.84), contrasted with near-zero IoU for pole (0.00) and traffic sign (0.05).

3.2. Fine-Tuning Results

All three strategies are trained for 8 epochs starting from EoMT-COCO. Table 1 summarizes mIoU on the Cityscapes validation set alongside the zero-shot lower bound (52.31%) and the EoMT-Cityscapes upper bound (81.68%). Figure 2 shows validation mIoU and training loss curves for all three strategies. All strategies converge without divergence (Figure 2b); the differentiated mIoU curves (Figure 2a) indicate that performance differences arise from adaptation scope rather than optimization instability. Strategy B peaks at epoch 4 and exhibits a slight decline in subsequent epochs, while Strategy C improves steadily throughout training. Strategy A (head-only) achieves 72.88% mIoU, a gain of +20.57pp over the zero-shot baseline. Strategy B reaches 74.31% but shows a mIoU decline after epoch 4, consistent with overfitting on the 2,975 Cityscapes training images. Strategy C achieves the best result of 75.54% with a stable training curve, surpassing Strategy B while using one-fifth of its trainable parameters.

3.3. Pixel-Based Baseline (ERFNet) and Post-Hoc Scoring

ERFNet (72.17% mIoU on Cityscapes) demonstrates relatively weak separating power between in-distribution and out-of-distribution (OOD) pixels. MSP yields poor performance, reaching an AuPRC of only 29.10 on SMIYC RA-21 and 12.42 on Road Anomaly, alongside FPR_{95} exceeding 62% on both benchmarks. MaxLogit emerges as the consistently strongest method for ERFNet, increasing AuPRC on RA-21 to 38.32 and on Road Anomaly to 15.58, while reducing FPR_{95} from 82.58% to 73.25% on RA. MaxEntropy metrics systematically fall between the MSP and MaxLogit baselines.

Table 1. Fine-tuning results on Cityscapes validation set (8 epochs). Lower bound: EoMT-COCO zero-shot (52.31%). Upper bound: EoMT-Cityscapes native (81.68%).

Strategy	Params	mIoU (%)
A: Head-only	∼6.7M	72.88
B: Blocks 8–11	∼36M	74.31
C: LoRA (r=8)	∼7M	**75.54**

3.4. Mask-Based Architectural Shift (EoMT)

Transitioning to the mask-based EoMT architecture produces a uniform performance leap across all evaluation metrics, benchmarks, and checkpoints. EoMT-Cityscapes increases the RA-21 AuPRC from ERFNet's 29.10 to 68.15. On SMIYC RO-21, it achieves a peak AuPRC of 94.05 with an exceptionally low FPR_{95} of 0.38%, validating the superior capability of mask-classification formulations in isolating anomaly boundaries. Across the various EoMT checkpoints, the relative performance ranking of the post-hoc methods remains highly stable. Conversely, RbA proves to be the least reliable metric overall.

3.5. Impact of Fine-Tuning and Temperature Scaling

The influence of domain-specific adaptation via LoRA is most visible on the Fishyscapes benchmarks. On the FS Lost & Found dataset, the LoRA-adjusted checkpoint increases the AuPRC to 47.86 (compared to 16.28 for EoMT-Cityscapes and 9.35 for EoMT-COCO). Similarly, it secures an AuPRC of 80.77 on FS Static, confirming that adaptation to local road geometries significantly improves OOD detection capabilities.

Finally, post-hoc confidence calibration via temperature scaling (MSP@best$T$) yields negligible gains. The absolute metric improvements never exceed 0.5% AuPRC, and consequently, FPR_{95} rates remain virtually unchanged.

4. Discussion

The zero-shot performance gap (52.31% vs. 81.68% mIoU) is primarily a semantic gap rather than a visual one: the DINOv2 backbone already encodes the visual knowledge required for road scene understanding, but the queries and

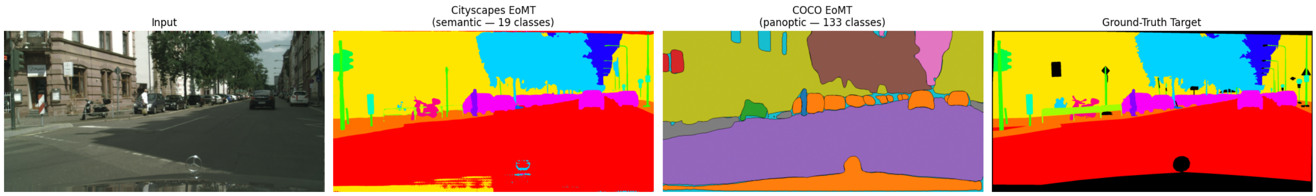
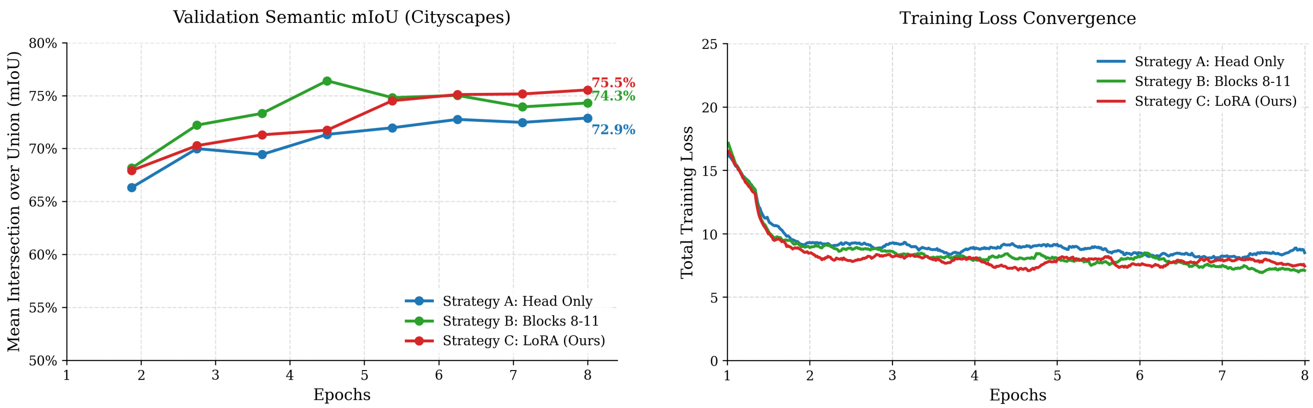


Figure 1. Qualitative comparison on a Cityscapes validation sample. From left to right: input image, EoMT-Cityscapes prediction (19 semantic classes), EoMT-COCO prediction (133 panoptic classes, after mapping), and ground-truth annotation. EoMT-Cityscapes produces a segmentation closely aligned with the ground truth, while EoMT-COCO struggles with domain-specific classes such as poles and traffic signs.



(a) Validation mIoU per epoch.

(b) Training loss convergence.

Figure 2. Training dynamics of the three fine-tuning strategies over 8 epochs. Strategy B (Blocks 8–11) peaks at epoch 4 and slightly declines, consistent with overfitting on the small Cityscapes training set. Strategy C (LoRA) converges smoothly to the best final mIoU of 75.5%, while Strategy A (Head-only) shows stable but slower improvement throughout training.

prediction heads are optimized for 133 COCO categories rather than 19 Cityscapes classes. Strategy A's +20.57pp gain from updating only the heads confirms this interpretation. Strategy C (LoRA) surpasses Strategy B despite using one-fifth of its trainable parameters because it applies a consistent adaptation signal across all 12 backbone blocks while the low rank ($r = 8$) provides implicit regularization, avoiding the overfitting observed in Strategy B's mIoU decline after epoch 4.

4.1. Anomaly Detection

Among the post-hoc scoring functions applied to ERFNet, MaxLogit consistently outperforms both MSP and MaxEntropy. The reason is straightforward: softmax normalization can produce high confidence values even when all raw logits are uniformly low (a known failure mode), whereas MaxLogit directly reads the raw magnitude of the winning class on unknown objects, no class produces a strong activation, keeping the maximum logit low and the anomaly score high. MaxEntropy improves marginally over MSP by considering the full class distribution rather than only the peak value. Regardless of the scoring method, ERFNet's absolute performance remains low across all benchmarks, confirming that per-pixel architectures lack the representational structure needed for reliable anomaly detection. Switching to the mask-based EoMT architecture yields a consistent performance jump. On EoMT, the gap between softmax-based methods narrows considerably; MaxEntropy emerges as the most reliable choice, likely because entropy captures more information from the composite quantity L than the peak alone. RbA proves to be the least stable method. It performs well in specific configurations, for instance, EoMT-Cityscapes on FS Lost&Found achieves a FPR95 of 9.31% - but degrades sharply in others (EoMT-COCO on RA-21 drops to 53.57% AuPRC versus ≈70% for standard methods). Since all models are evaluated on the same 19-class space after remapping, this instability does not stem from class cardinality directly. Rather, RbA's sum-of-sigmoids formulation is sensitive to the absolute scale and distribution of L: when L values result from marginalizing 133 COCO classes down to 19, probability mass is lost in the aggregation, producing noisier sigmoid responses across all channels and weakening the "rejected by all" signal. A similar effect occurs with the LoRA checkpoint, where

Table 2. Anomaly segmentation results on five benchmarks and semantic segmentation mIoU on Cityscapes. AuPRC (higher is better) and FPR95 (lower is better) are reported in %. mIoU is on the Cityscapes validation set. MSP@bestT uses the best temperature T per dataset. The full temperature grid (T 0.5, 0.75, 1.0, 1.25, 1.5, 2.0) is available in the supplementary material; the best T per dataset is reported here.

			SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
Model	**mIoU**	**Method**	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
ERFNet	72.17	MSP	29.10	62.55	2.71	65.23	1.75	50.60	7.47	41.84	12.42	82.58
		MaxLogit	38.32	59.34	4.63	48.44	3.30	45.49	9.50	40.30	15.58	73.25
		MaxEntropy	30.97	62.66	3.04	65.91	2.58	50.16	8.84	41.55	12.67	82.75
EoMT-Cityscapes	81.68	MSP	68.15	30.20	94.05	0.38	16.28	12.92	58.07	41.30	71.31	15.48
		MaxLogit	67.62	32.34	94.08	0.37	16.26	12.73	58.16	47.84	70.78	15.11
		MaxEntropy	68.31	30.48	94.13	0.36	18.64	12.72	56.83	41.20	73.61	14.72
		RbA	64.47	96.80	93.86	0.37	16.40	9.31	60.17	78.83	71.01	14.24
		MSP@bestT	68.34	30.17	94.21	0.37	16.35	13.85	58.21	41.15	71.55	15.43
EoMT-COCO	52.31	MSP	70.60	49.97	90.04	13.00	9.35	90.35	85.19	11.23	77.10	74.54
		MaxLogit	71.01	48.18	90.65	11.36	9.38	90.62	85.53	10.42	76.67	73.05
		MaxEntropy	71.91	33.15	90.05	16.53	9.02	90.72	86.08	9.56	76.39	75.03
		RbA	53.57	68.59	88.01	43.70	4.43	97.12	85.77	52.75	73.18	59.39
		MSP@bestT	70.77	49.67	90.06	13.00	9.56	90.26	85.17	11.03	77.08	74.79
EoMT-FineTuned-LoRA	75.54	MSP	71.05	99.67	92.80	0.89	47.86	19.06	80.77	9.65	67.35	24.99
		MaxLogit	70.16	99.66	92.90	0.84	47.90	19.67	80.89	9.50	67.43	24.39
		MaxEntropy	68.83	99.66	93.14	0.76	47.95	18.65	81.34	8.97	69.34	23.03
		RbA	55.22	99.28	92.89	0.68	46.81	82.70	79.71	85.19	67.01	18.74
		MSP@bestT	71.11	99.67	92.72	0.92	48.13	19.03	80.73	9.62	67.42	24.71

fine-tuning inflates L values on driving scenes, saturating sigmoid activations and reducing discriminability. Unlike MSP or MaxEntropy - which depend only on the relative ranking of pixel scores - RbA's behavior depends on absolute activation levels, making it inherently less robust across model configurations.

4.2. Checkpoint Profiles and Temperature Scaling

The evaluated EoMT checkpoints display distinct, complementary profiles highlighting an in-domain vs. OOD robustness trade-off. EoMT-COCO excels when anomalies are generic, COCO-like objects (FS Static AuPRC up to 86%), but performs poorly on FS Lost&Found (AuPRC $\approx$ 9%) and shows high FPR95 on RA-21 and Road Anomaly, as the network struggles with background variations deviating from standard driving environments. EoMT-Cityscapes demonstrates robust performance on structured driving scenes (RO-21 AuPRC 94%, FPR95 $\approx$ 0.37%) and a balanced profile on Road Anomaly. EoMT-FineTuned-LoRA delivers the largest gain on FS Lost&Found (AuPRC from $\approx$16% to $\approx$48%) and improves FS Static, since fine-tuning specializes the model for typical road obstacles. The downside is a pronounced FPR95 increase on RA-21 ($\approx$99.7%): fine-tuning makes the model overly confident on driving scenes, destroying its ability to distinguish normal background from anomalies at the 95% TPR threshold.

Finally, temperature scaling yields marginal returns. While grid search identifies an optimal $T \approx 2.0$, the resulting calibration does not fundamentally alter the underlying pixel ranking. This indicates that in mask-transformer architectures, the spatial ranking of anomalies is largely predetermined by the structure of L, confirming that while calibration is implemented correctly (as evidenced by shifting values at $T \neq 1$), its impact on post-hoc ranking metrics is negligible.

5. Conclusions

This work investigated road scene understanding through the lens of mask-based segmentation and anomaly detection. Our zero-shot evaluation confirmed that the performance gap between COCO and Cityscapes checkpoints is semantic rather than visual, with the DINOv2 backbone already encoding sufficient visual knowledge for driving scenes. Among the fine-tuning strategies, LoRA achieved the best mIoU (75.54%) with only 7M trainable parameters, demonstrating that low-rank adaptation provides an effective balance between expressiveness and regularization on small domain-specific datasets. For anomaly segmentation, mask-based architectures consistently achieved higher AuPRC than the per-pixel ERFNet baseline across all benchmarks and scoring methods, though specific checkpoint–dataset combinations exhibited elevated FPR95 due to overconfidence. MaxEntropy proved the most robust post-hoc method on EoMT, while RbA showed high sensitivity to the underlying distribution of L across model configurations. Temperature scaling yielded negligible improvements, suggesting that the spatial ranking of anomalies is largely predetermined by the mask-query structure of L. Our work was limited to ViT-B scale evaluation due to computational constraints. Future directions include scaling to ViT-L, integrating learned OOD objectives such as outlier exposure, and extending evaluation to video sequences for temporal consistency.

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For the development of the project code, Claude Sonnet 4.6 (Anthropic) was used as a debugging assistant. Specifically, the tool was consulted to identify and resolve implementation errors during development. All architectural decisions, experimental design, hyperparameter choices, and result analysis were performed exclusively by the authors. The use of the tool was limited to resolving specific technical issues under continuous human supervision, with all proposed fixes reviewed and validated by the authors before integration.

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